Tutorial: Output Stage Design

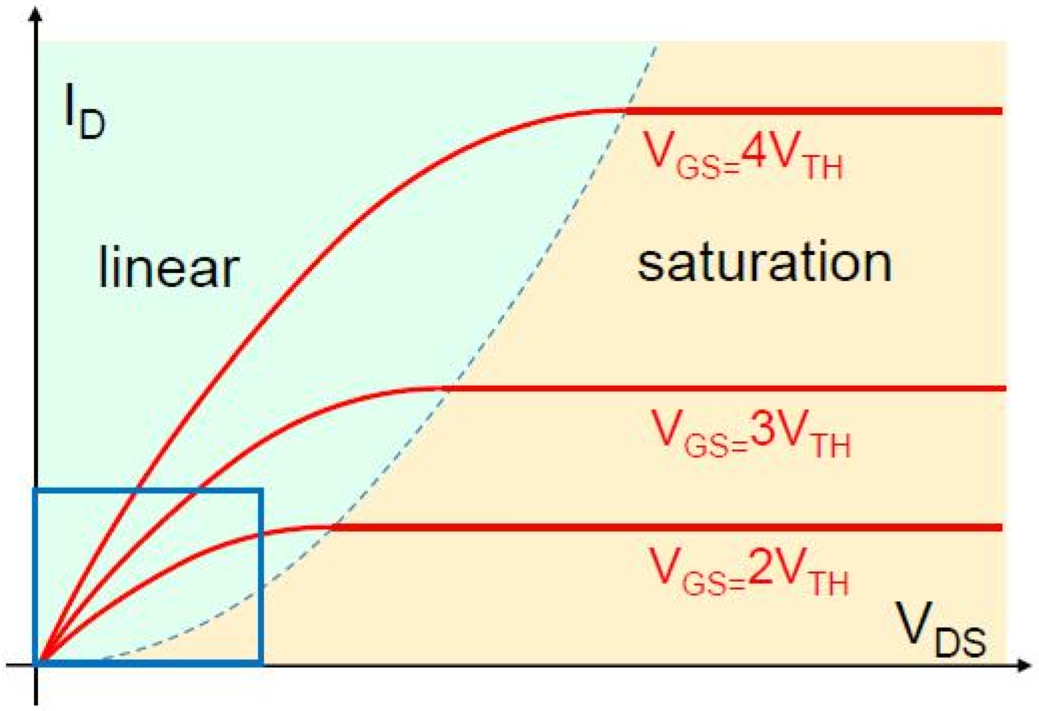
ET4382 Q3 2021/22

Agenda

- Finding R_{ON} and Q_G using simulation
- Simulation of power stage

Power FETs

- Conduction Loss: I^2R_{ON}
- How to determine R_{ON}?
- R = V/I
  - Power FET fully ON
  - Usually take V=100mV
- Temperature-dependent



[Lecture 1 Slides]

Power FETs

- Gate Charge Q_G

$$Q_{GL} = C_{GSL} \cdot (V_{REG} + V_{SUP}) + C_{GDL} \cdot (V_{REG} + 3V_{SUP})$$

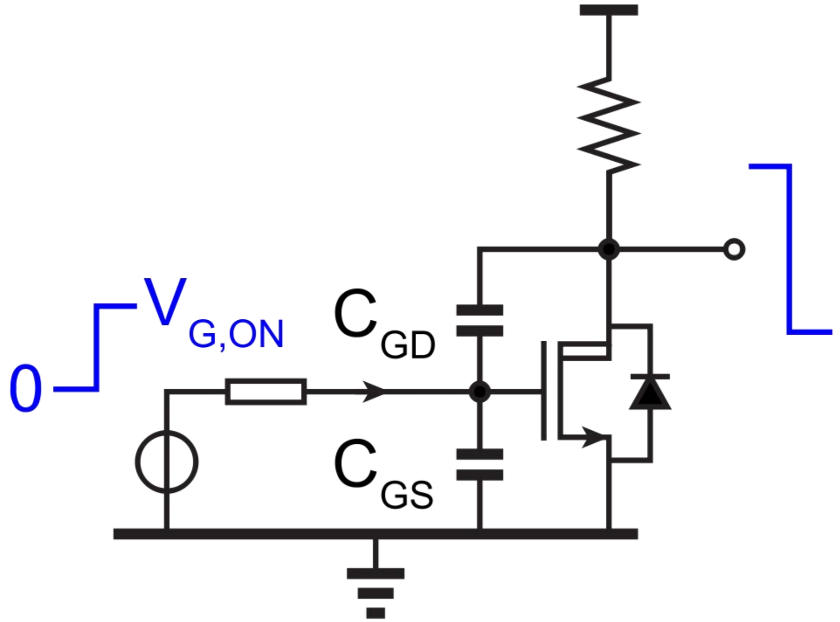
Miller effect

$$Q_{GH} = Q_{GL}$$

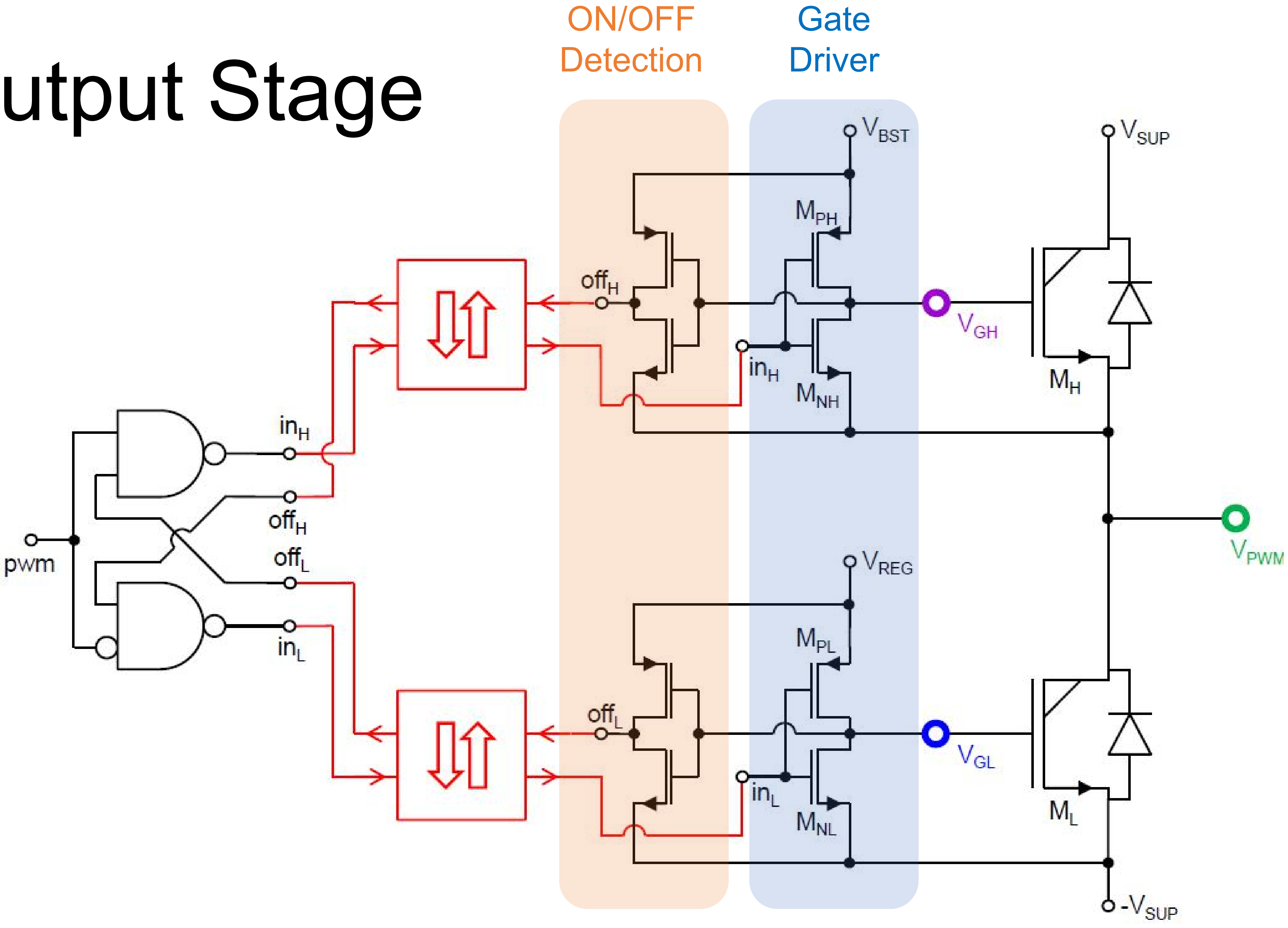
- Gate Charge supplied by $+V_{SUP}$

$$P_Q = f_{PWM} \cdot 2V_{SUP} \cdot (Q_{GL} + Q_{GH})$$

- How to determine Q_G?
  - C_{GS} and C_{GD} are nonlinear
- Simulation
  - Should include Miller effect
  - *integ/iinteg* function



Output Stage



[Lecture 3 Slides]